The Definition of the Derivative

ANSWER KEY

Section 1: Difference Quotients & Limit Definitions

1 Which limit expression represents the derivative $f'(x)$ by definition?

a. $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$ ✓

b. $\lim_{h\to 0} \frac{f(x+h)+f(x)}{h}$

c. $\lim_{h\to \infty} \frac{f(x+h)-f(x)}{h}$

d. $\frac{f(x+h)-f(x)}{h}$

By definition, $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$.

2 Evaluate $\lim_{h\to 0} \frac{(3+h)^2-9}{h}$.

a. 3

b. 6 ✓

c. 9

d. 0

This limit represents $f'(3)$ for $f(x) = x^2$. $f'(x) = 2x \Rightarrow f'(3) = 2(3) = 6$.

3 Which function and value of c are represented by $\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4}$?

a. $f(x) = \sqrt{x}$ **at** $c = 4$ ✓

b. $f(x) = x^2$ at $c = 2$

c. $f(x) = \sqrt{x}$ at $c = 2$

d. $f(x) = \frac{1}{\sqrt{x}}$ at $c = 4$

Using the alternate form $f'(c) = \lim_{x\to c} \frac{f(x)-f(c)}{x-c}$, this corresponds to $f(x) = \sqrt{x}$ at $c = 4$.

Section 2: Differentiability Implies Continuity

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4 If a function f is differentiable at $x = c$, which of the following statements MUST be true?

a. f **is continuous at** $x = c$ ✓

b. $f'(c) > 0$

c. $f''(c)$ exists

d. f has a local extremum at $x = c$

Differentiability implies continuity. (The converse is false, as demonstrated by sharp corners like $f(x) = |x|$).

5 Why is the function $f(x) = |x - 3|$ not differentiable at $x = 3$?

a. **The left-hand and right-hand derivative limits are unequal (corner)** ✓

b. $f(3)$ is undefined

c. f is not continuous at $x = 3$

d. $\lim_{x \to 3} f(x) = \infty$

For $x < 3$, $f'(x) = -1$, and for $x > 3$, $f'(x) = 1$. The corner creates unequal one-sided derivative limits.

6 The graph of $f(x) = x^{1/3}$ has what feature at $x = 0$ that prevents differentiability?

a. **A vertical tangent line where** $f'(x) \to \infty$ ✓

b. A jump discontinuity

c. A sharp cusp

d. A hole

$f'(x) = \frac{1}{3}x^{-2/3} = \frac{1}{3x^{2/3}}$. As $x \to 0$, $f'(x) \to \infty$, which corresponds to a vertical tangent line.

Section 3: Tangent and Normal Lines

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7 What is the equation of the tangent line to $f(x) = x^3 - 2x$ at the point $(2, 4)$?

a. $y - 4 = 10(x - 2)$ ✓

b. $y - 4 = 8(x - 2)$

c. $y - 4 = 12(x - 2)$

d. $y = 10x - 12$

$f'(x) = 3x^2 - 2$. Slope at $x = 2$ is $f'(2) = 3(4) - 2 = 10$. In point-slope form: $y - 4 = 10(x - 2)$.

8 What is the slope of the normal line to $y = x^2 + 3x$ at $x = 1$?

a. $-\frac{1}{5}$ ✓

b. 5

c. -5

d. $\frac{1}{5}$

$y' = 2x + 3 \Rightarrow y'(1) = 5$. The normal line is perpendicular to the tangent line, so its slope is $-\frac{1}{5}$.

9 At what point on the parabola $y = x^2 - 4x + 3$ is the tangent line horizontal?

a. $(2, -1)$ ✓

b. $(2, 0)$

c. $(4, 3)$

d. $(0, 3)$

Horizontal tangent requires $y' = 0$: $2x - 4 = 0 \Rightarrow x = 2$. $y(2) = 4 - 8 + 3 = -1$. Point is $(2, -1)$.

Section 4: Estimating Derivatives from Tables & Graphs

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10 Given values $t = 2, 4, 6$ with $s(2) = 14, s(4) = 30, s(6) = 54$, estimate $s'(4)$ using the average rate of change over $[2, 6]$.

a. 10 ✓

b. 8

c. 16

d. 12

Symmetric difference quotient: $s'(4) \approx \frac{s(6) - s(2)}{6 - 2} = \frac{54 - 14}{4} = \frac{40}{4} = 10$.

11 If the graph of f has a local maximum at $x = 5$ and is differentiable, what is $f'(5)$?

a. 0 ✓

b. 1

c. Positive

d. Undefined

By Fermat's theorem, the derivative at an interior local extremum of a differentiable function is zero: $f'(5) = 0$.

12 If $f'(x) > 0$ for all $x \in (a, b)$, what does this indicate about the graph of f on (a, b)?

a. f **is strictly increasing** ✓

b. f is strictly decreasing

c. f is concave up

d. f is linear

A strictly positive first derivative implies that the function is strictly increasing on the interval.

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